

Agentic AI

AI Agent

Chat bot $\Rightarrow$ Answers what you ask

Workflow $\Rightarrow$ Follows a predefined sequence.

AI Agent $\Rightarrow$ Thinks & decides next steps dynamically.

* AI Agent (LLM-based) = LLM + Tools + loop.

Characteristics

$\rightarrow$ Goal Directed = works toward an objective.

$\rightarrow$ Autonomous = Decides what to do next without being told at each step.

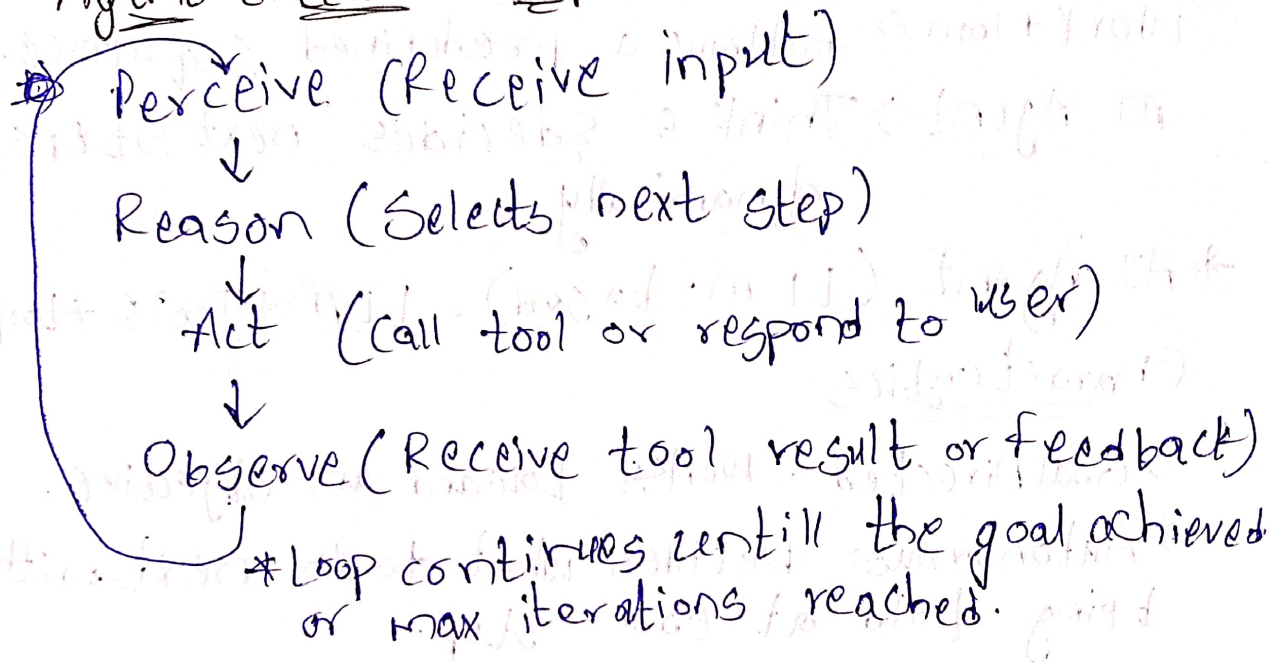
$\rightarrow$ Tool-using = Interacts with API's, databases, code execution

$\rightarrow$ Iterative = Observe $\rightarrow$ Reason $\rightarrow$ Act $\rightarrow$ Observe Again.

Capability	Chatbot	AI Agent
Knowledge Access	Training data & provided context (prompt)	Training + live tools
Actions	Generate text	Search, compute, call API's & write code.
Memory	None between calls	Working + persistent memory.
Planning	Single-pass response	Multi-step planning & iteration.
Self-Correction	Can do internal consistency checks	Observes results.

* AI Agent is an architecture pattern that wraps an LLM in a reason-act-observe loop.

Agent Execution Loop



AI Agent Core Components

→ LLM

→ Tools (Bridge between the agent & external world).

→ Loop (Orchestration).

1) LLM (Agent's Brain)

- Understands user intent from natural language
- Plans multi-step sequences to achieve the goal.
- Decides which tool to call & with what arguments.

* The LLM is the reasoning core.

2) Tools (Agent's Hands)

Agent Defines Tools → Agent sends Tools to LLM → LLM Decides

* LLM never execute tools directly.

* Tools transform a language model from a text generator into an actor capable of reading info & executing actions.

↓
Agent Executes Tool

↓
Agent Returns results to LLM.

3) Loop (Agent's Nervous System)

→ Loop Management

→ Reasoning Strategy

→ Memory Management

→ Tool Routing

→ State Machine

→ Safety & Guardrails

* Decides when to reason, which tool to call & how results inform the next step.

Reasoning Patterns

→ CoT (Chain of Thought)

* Thinking Step by Step.

→ ReAct (Reasoning + Acting)

Agent Framework

1) Langchain/Langgraph
→ Largest ecosystem, graph-based stateful workflows.

2) OpenAI Agents SDK

* Define agents & orchestrate multi-agent workflows.

3) CrewAI

4) Hugging Face SmolAgents

Safety & Guardrails

→ AI Agent Threat Model

1) Prompt Injection ⇒ Attacker hijacks AI Agent via crafted input or poisoned retrieved context.

2) Tool Misuse ⇒ Agents can call tools with wrong or dangerous arguments.

3) Memory Poisoning ⇒ Poisoned content stored in memory can affect future behavior of the agent.

4) Data Exfiltration ⇒ AI Agents can be tricked into leaking sensitive internal data through tool calls or responses.

5) Resource Exhaustion ⇒ Infinite loops or excessive API calls cause massive cost overruns.

Defense in Depth

1.) Input Validation

- * Treat all external/retrieved content as untrusted data.

2.) LLM Guardrails

- * Safety system prompts, tool access controls

3.) Tool Boundaries

- * Each tool has min access permissions, sandboxing.

4.) Output Filtering

- * Content policy, relevance verification.

5.) Observability

- * Checks inputs, tool calls, outputs, errors, costs.